Introduction to Stata 7.0: Economics 311[1]

I. Access to Stata

Stata is in a folder on your dock titled "IRC Applications." Stata can be accessed from other machines on campus by selecting the "Data Analysis and Processing" folder in the Academic Server.

Stata is a "keyed" program, so you need to be on campus to use the program.

II. Starting Stata

Double-click on the file titled "**Stata**" in the **IRC Applications** folder.

A. Stata Windows

- review
- results
- variables
- command

B. Stata Toolbar

13 buttons – bring your mouse over a button and a box will appear with a description of that button.

C. Stata Log File

A log file is a record of your Stata session. Log files can either be in a Stata format (SMCL) or a text (ASCII) format. Saving the log file as a text file will allow you to bring the file into *Word* for additional editing.

Start a log file by clicking on the Log button, select begin, and fill in a filename.

You can add comments to your log by typing a star (*) at the beginning of a command line. This will treat that line as a comment.

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[1] This handout draws liberally from Stata 7: Getting Started, Macintosh. 2001. College Station, TX: Stata Corporation.

III. Stata's Help Feature

Choosing **Help** from the menu allows you to:

1. See the help table of contents
2. Search for help entries on a topic
3. Get help for a Stata command

Choosing **Search...** from the Help menu allows you to enter keywords and produces a screen with hypertext links (in blue) that will take you to the help files for the appropriate Stata commands. You will also see references to the topic in the Reference Manual, Graphics Manual, User's Guide, etc.

> Example:
>
> Select **Search** from the Help menu
> Enter regression and click **OK**
> Scroll down to **regress** and click on this word
>
> *Use proper English and statistical terminology with **Search**

Choosing **Help Contents** from the Help menu gives a list of Stata's help table of contents. You can:

1. Choose from the links on this page to view help for a particular command
2. Or enter the full name of the Stata command in the edit field at the top of the Help window.

> Example:
>
> 1. Type **ttest** and press Enter
>
> *Only enter Stata commands – using proper English or statistical terminology will probably not work

The help files contain a lot of information, but not as much as the *Reference Manual*, *Graphics Manual*, and *User's Guide*. These publications are on reserve at the Reed Library and in the Public Policy Workshop.

Help will let you know where to find more information about specific topics in these manuals. For example,

"[U] 2.4 **The Stata Technical Bulletin**" means section 2.4 in the *User's Guide*.
"[R] **regress**" means the entry **regress** in the *Reference Manual*.
"[G] **graph options**" means the entry **graph options** in the *Graphics Manual*.

Example:

1. Choose **Help** from the menu bar and select **Search...**
2. Enter data and click **OK**
3. Scroll down until you see [R] describe. **describe** is a Stata command that describes the contents of data in memory or on disk. The [R] means that documentation is in the *Reference Manual*. An on-line help file exists for this command.
4. Click on the hypertext link "describe" in "help describe."
5. The help file for Stata commands contain:
    - The command's syntax
    - A description of the command
    - Options
    - Examples, and
    - References to related commands.

IV. Inputting Data into Stata using the **Data Editor**

Click on the **Data Editor** button or type edit and press Return in the Command window.

Stata's editor looks like a spreadsheet and it functions in a way that is quite similar to *Excel*.

A. Inputting Data

Things to know about entering data in Stata

- Quotes around string variables are unnecessary
- A period ('.') represents a missing numeric value
- Press Tab or Return to input a missing numeric value
- Press Tab or Return to input a missing value for a string variable
- Stata will not allow empty columns or rows in the middle of your dataset

Example:
1. Enter the auto data on the Session 1 handout into Stata's editor. You can do this variable-by-variable or observation-by observation.
2. When entering data observation-by-observation use the tab key. Stata's tab key is smart. Notice what happens after you've entered the first observation.

B. Renaming Variables

Double-click anywhere in the variable's column. This brings up the Variable Information dialog box. Enter the new name of the variable. Label allows you to specify a more detailed description of the variable.

Rules for variable names:

- Stata is case sensitive
- A variable name must be between 1 and 8 characters long
- Characters can be letters, digits, or underscores
- Spaces or other characters are not allowed
- The first character of a variable name must be a letter or an underscore

C. Copying and Pasting Data

1. Select the data you want to copy

   Click and drag the mouse to select a range of cells

2. Copy the data to the clipboard

   Pull down on the **Edit** menu and choose **Copy**

3. Paste the data from the clipboard

   Click on the top left cell of the area to which you wish to paste. Pull down the **Edit** menu and choose **Paste**.

D. Exiting the Data Editor

Click on the editor's close box.
Changes that you made in the editor are not saved until you tell Stata to save them. Data can be saved by pulling down **File** and choosing **Save As**.

You cannot save your data until you have exited the editor.

Example:

1. Click on the **File** menu and select **Save**.
2. Enter the filename **afewcars** Stata will automatically add the .dta extension to the file.
3. Type **clear** in the Stata command window. This removes the dataset from Stata's memory.

V. Inputting Data from a File

A. Insheet

The insheet command is used to import text (ascii) files created by a spreadsheet program. It is important that the file be saved in the spreadsheet program as "text only" with a tab or comma column delimiter. The general format of the insheet command is:

insheet using "filename"

If the file is not in the current folder type "insheet using" then select Filename from the File menu and select the file.

Example:

1. Import the file "SavingsIncome-UK.txt" (a tab delimited text file) from the Econ 311 folder.
2. Type **browse** in your command window. This allows you to view, but not change the data. Exit the browser.
3. Type **clear** in the Stata command window.

VI. Labeling Data

Using the dataset **afewcars.dta**

Example:

1. Type **use afewcars** into the Stata command window
2. Type **describe** into the Stata command window

The data description provides information on the variable name, storage type, and display format.

Example:

1. Type clear in the Stata command window.
2. Open the file "auto.dta" in the Econ 311 folder.
3. Use the **describe** command

VII. Editor/Browser

A. Editor

The editor has several buttons:

Preserve
Restore
Sort
<<
>>
Hide
Delete

Example:

1. Using the auto.dta file
2. Open the data editor
3. Use the sort button to list cars based on their price
4. Use the ">>" key to move the "weight" variable so it is next to the "make" variable.
5. Delete the "trunk" variable.
6. Make other changes to the data.
7. Click on restore. The changes that you have made have been reversed.
8. Exit the editor.
9. Look at the Stata Results window. This has recorded the changes that you have made.

B. Browse

Click on the Data Browser button or type browse in the Command window. This allows you to view your data, but not to change it.

Example:

1. In the command window type

```
browse make mpg price if foreign == 1
```

This displays the make, mpg, and price of those cars that are designated as "foreign" in the data set.

VIII. Shortcuts!

A. Review Window

Click on a command in the Review Window and it is copied into the Command Window.

> Example:
>
> 1. Using the file auto.dta, type regress mpg weight in the Command Window. Press return.
>
> 2. Click on this command in the Review Window and add the variable foreign. Press return.

Double-clicking on a command in the Review Window executes the command.

The Review Window is handy if you've made a mistake and need to fix a typo.

B. Variable Window

Clicking on a variable name copies it into the Command Window.

C. Function Keys

Some of the F-keys are defined to have special meanings:

F3: Describe
F7: Save

VIII. Listing Data

A. List

Typing list in the Command Window lists the entire data set. A subset of variables can be listed.

> Example:
>
> 1. Type list make mpg price in the Command Window.

B. List with in

The Stata command **in** restricts the list to a range of observations

Positive numbers count from the top of the data. Negative numbers count from the end of the data

You can specify both a variable range and an observation range.

Example:

Type the following commands in the Command Window using the file "auto.dta"

1. list
2. list in 1
3. list in –1
4. list in 2/4
5. list make mpg in –3/-2

C. List with if

The Stata command **if** restricts the observations to meet certain criteria using logical operators. The logical operators are:

<	less than
<=	less than or equal
==	equal
>=	greater than or equal
>	greater than
~=	not equal (~! can also be used)
&	and
|	or
~	not (! can also be used)
()	parentheses specify order of evaluation

Example:

1. list
2. list if mpg > 22
3. list if mpg > 22 & mpg ~=.
4. list make mpg if mpg> 22 | (price > 8000 & gear_ratio > 3.5)
5. list make mpg if mpg > 22 | (price > 8000 & gear_ratio > 3.5) in 1/4

Notes:

1. Tests of equality are specified with double equal signs (==)
2. Joint tests are specified with an &, not multiple ifs.
3. Tests with strings are allowed, but the contents of the string variable must be enclosed in double quotes: if make == "AMC Concord."

IX. Creating New Variables

A. Generate

Generate allows you to create a new variable that is an algebraic expression of other variables. Generate can be abbreviated by the letter "g" or the term "gen."

Example: Using the data set auto.dta

1. gen logpr = ln(price)
2. gen ratio = price/mpg
3. gen silly = ((price+100)/ln(mpg-3))^2

B. Replace

The command replace allows you to change the content of existing variables.

Example:

1. replace weight = weight/1000

New variables can be created based on logical requirements about existing variables. This is handy when working with dummy variables. For example, suppose you want to create a new variable that is the predicted price of domestic and foreign cars for next year. Domestic cars are estimated to increase in price by 5% while foreign cars are expected to go up by 10%. The following commands will reflect these changes:

Example:

1. gen predpric = 1.05*price if foreign==0
2. *generates a new variable predpric and sets all observation values equal to zero.
3. replace predpric = 1.1*price if foreign == 1
4. list make weight price predpric foreign
5. *using the list command allows you to check your data to make sure the changes are correct.

X. Deleting Variables and Observations

A. Clear and Drop_All

The commands **clear** and **drop_all** eliminate data from memory. **drop_all** drops the data from memory. **clear** resets Stata.

B. Drop

The **drop** command allows you to drop variables and/or specific observations.

Example: Using auto.dta

```
1. drop in 1/3  *this drops observations 1 through 3
2. drop if mpg > 21
3. drop gear_ratio
4. *this drops the variable gratio
5. list
6. *this allows you to check your work
```

To make changes permanent, resave the data by choosing **Save** under the **File** menu.

XI. Working with data

A. Preliminaries – describe and list

When working with an unfamiliar data set it is useful to describe the data. The Stata command describe provides information on the number of observations, variables, variable type, etc.

More detailed information about the data set can be obtained using the Stata command **list**.

Example: Using auto.dta

```
1. describe
2. list
3. list make mpg in 1/10
4. sort mpg
5. *the sort command sorts from low to high
```

B. Descriptive Statistics

The Stata command **summarize** provides summary statistics of the data set. Logical operators can be combined with summarize.

Example:

1. summarize
2. summarize price if mpg < 21
3. summarize mpg, detail
4. *this provides percentiles, the median value, the four smallest and four largest values.

C. Tables

Frequency tables are obtained using the tabulate command.

Example:

1. tabulate foreign
2. *provides the frequency and percent of foreign and domestic cars
3. tabulate rep78 foreign
4. *provides frequency-of-repair records for foreign and domestic cars

D. Correlation Matrices

The correlation between variables is calculated using the Stata command correlate. Correlation matrices can contain multiple variables.

Example:

1. correlate mpg weight
2. correlate mpg weight if foreign == 0 *this calculates the correlation of weight and mpg for domestic cars
3. correlate

E. Graphing Data

The Stata command graph followed by the two variables will produce a scatterplot. Stata's graphing features are quite robust. For additional information see the *Stata Graphics Manual*.

Example:

1. sort foreign
2. graph mpg weight
3. graph mpg weight, by (foreign) total
4. *this produces three graphs – one showing the relationship between mpg and weight for domestic cars, another for foreign cars, and a third for the observations combined

F. Linear Regression

Based on the graph of mpg and weight which appears to be nonlinear, the following regression equation is hypothesized:

$$mpg = \beta_0 + \beta_1 weight + \beta_2 weight^2 + \beta_3 foreign$$

The weight^2 variable needs to be generated. Foreign is in the data set as a dummy variable.

Example:

1. gen wtsq = weight^2
2. regress mpg weight wtsq foreign
3. predict mpghat
4. *this post-estimation command gives the predicted values for the dependent variable (mpg). This will allow us to graph the predicted curve.
5. sort weight
6. *you need to sort the data by the x-variable before graphing so the points are connected in the right order.
7. graph mpg mpghat weight if foreign ==0, connect (.l) symbol (Oi)
8. graph mpg mpghat weight if foreign == 1, connect (.l) symbol (Oi)
9. Note: this instructs the program to graph mpg vs. weight and mpghat vs. weight. Connect (.l) tells Stata not to connect the mpg vs. weight points – this is the '.', but to connect with a straight line, the mpghat vs. weight points. Symbol (Oi) instructs Stata to use big circles for the mpg vs. weight points, but to use no symbol for the mpghat vs. weight points.